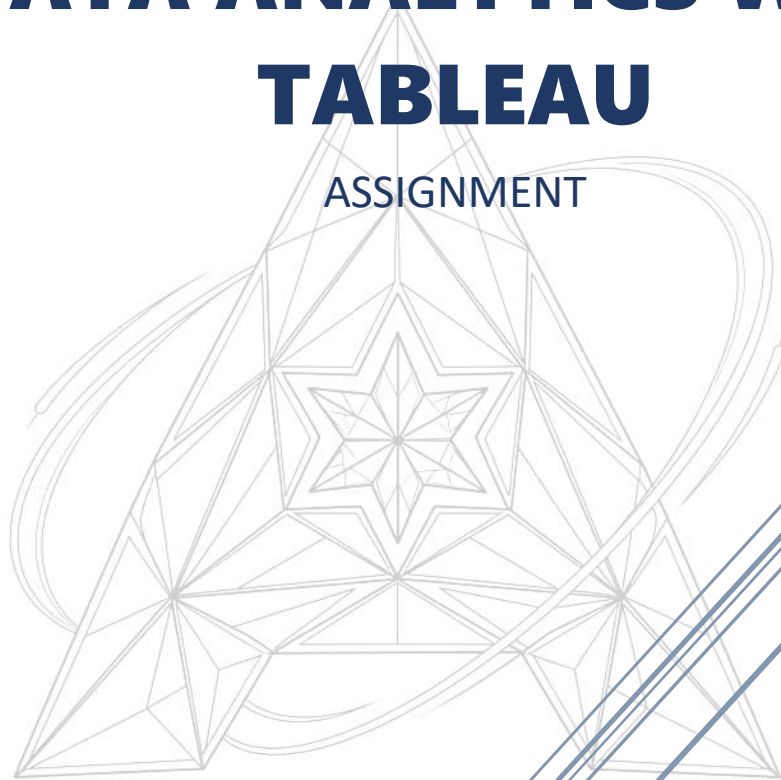


Name of the student: Polakala Akshaya

Hall Ticket No: 0425046033

DATA ANALYTICS WITH TABLEAU

ASSIGNMENT



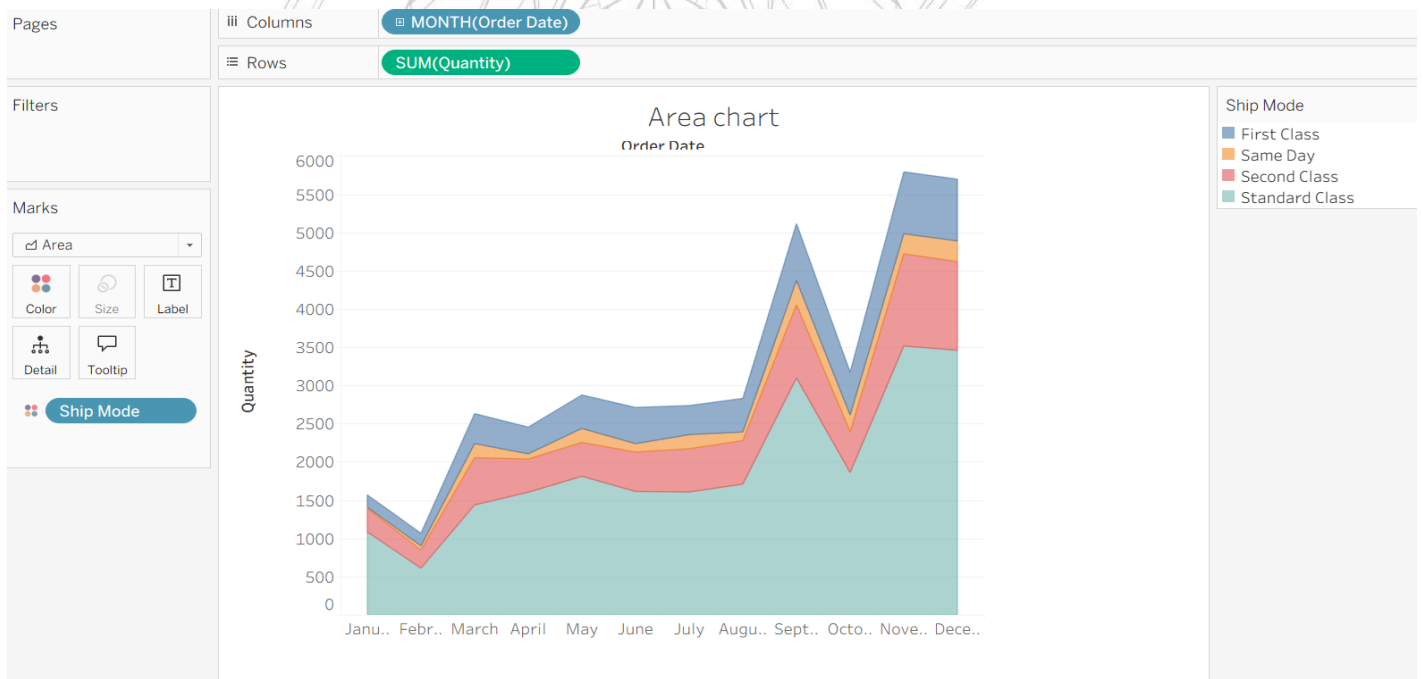
ALPHA
- INNOVATORS -



Area chart

1. From data pane drag **[Order Date]** into the Columns shelf
2. Right Click **[YEAR-ORDER DATE]** on column
3. Change the format from Year to **[Month]**.
4. Move **[Quantity]** into the Rows section.
5. Drag **[Ship Mode]** to the Color option in Marks.
6. In Marks card click on **(Automatic)** drop-down arrow
7. Select the chart type
8. Choose **Area**

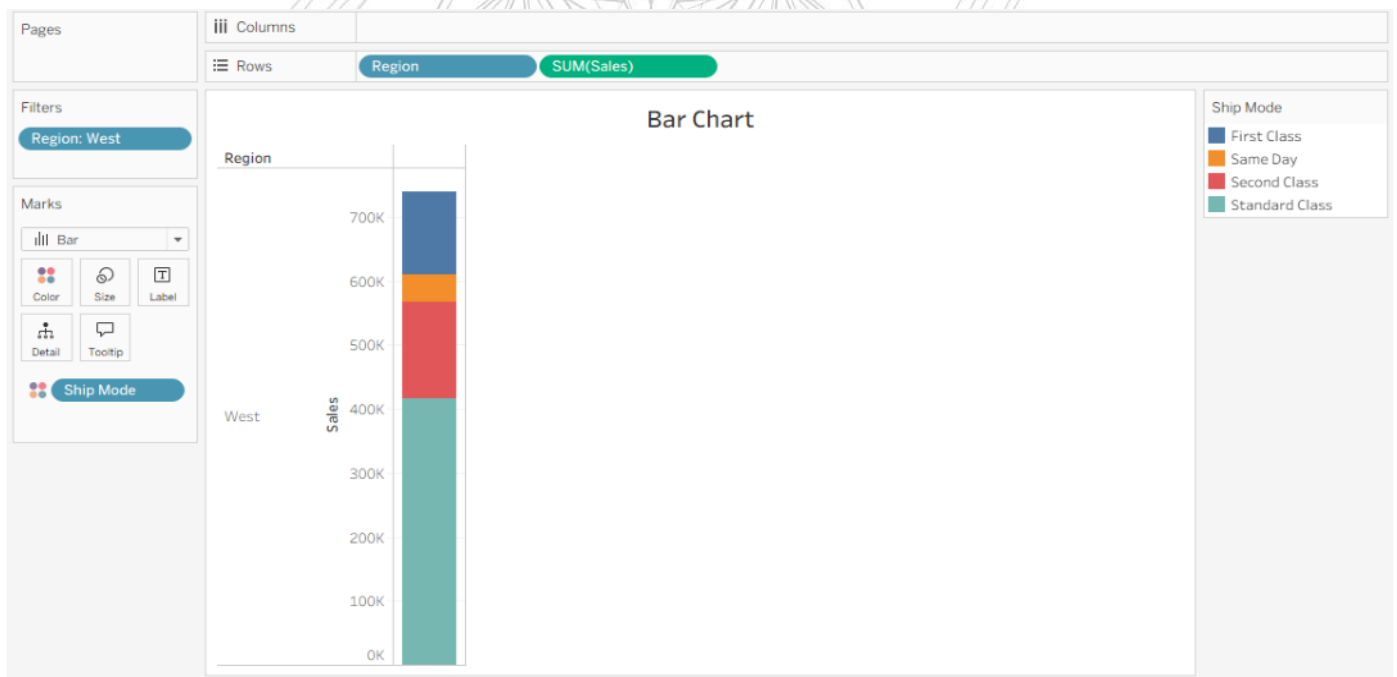
Output:



Bar chart

1. Add **[Sales]** to the row shelf
2. From Marks card menu click **Automatic drop-down arrow**
3. select **Bar chart type**
4. Place **[Ship Mode]** in the color area
5. Move **(Region)** into the Rows shelf placed in left side of ship mode
6. Again, drag **(Region)** into the filters shelf
7. Un check all the option keep only the **[WEST Region]**
8. Press ok the bar chart generates on work place

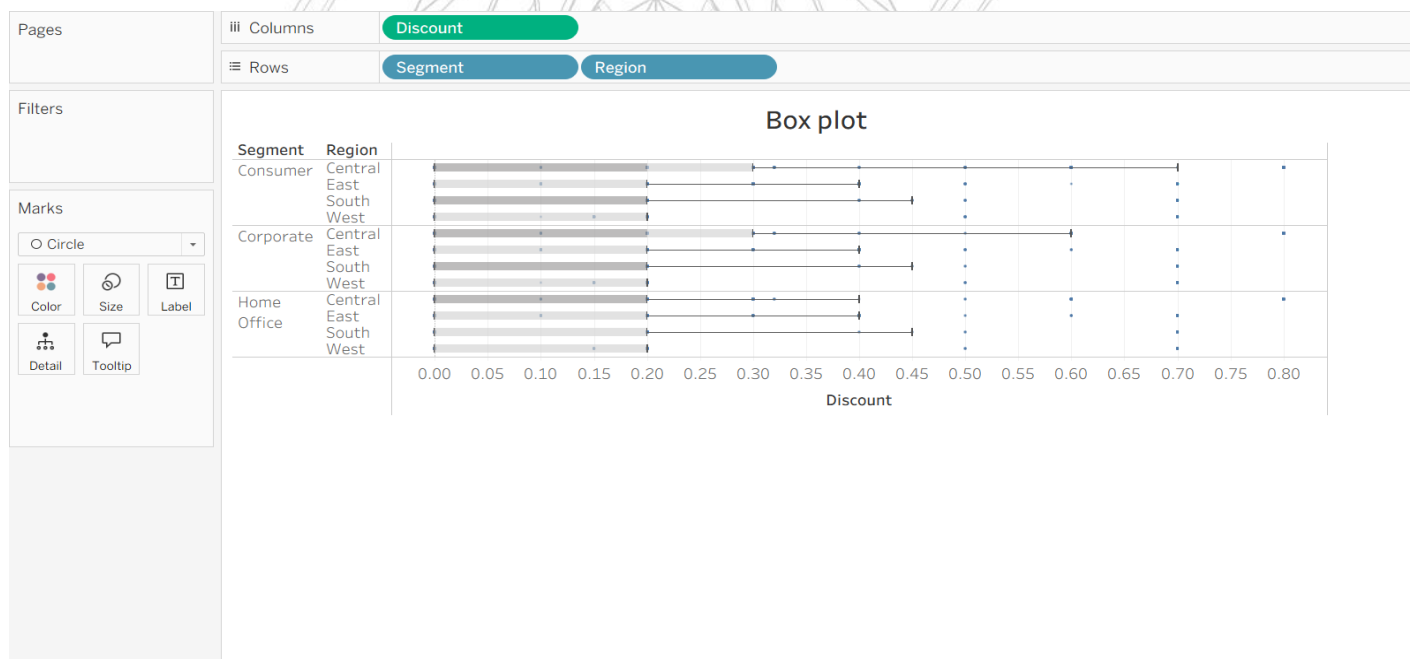
Output:



Box plot

1. Drag **[Segment]** into the columns shelf
2. Drag **[Discount]** into rows shelf
3. Right click on **[SUM (Discount)]**
4. click on **Measures (sum)** and select Average
5. Drag **[Region]** into right side of segment in columns shelf
6. Top right corner Click show me
7. Select **Box-and-whisker plot**
8. Move **Region** from marks card back to column shelf place on right side of segment
9. Click **Analysis Menu** in toolbar
10. Uncheck **Aggregate measures**
11. Click swap Rows and columns button using ctrl+w short key

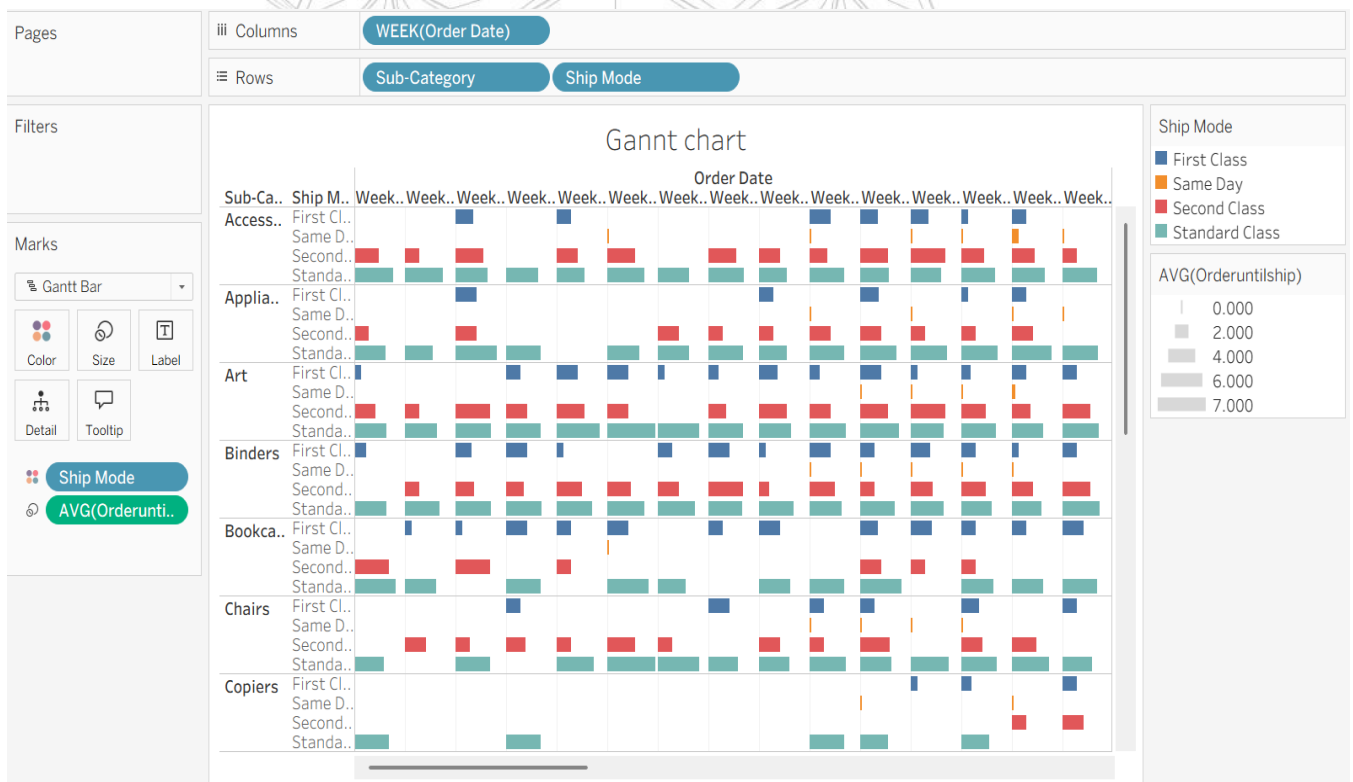
Output:



Gantt Chart

1. Drag **[Order Date]** placed into Column shelf
2. Move **[sub-category]** placed into Rows shelf
3. Right click on **[YEAR ORDER DATE]** in column shelf select year to **[week number]** format
4. Move **[Ship Mode]** into Rows shelf placed in right side sub-category
5. In Mark card click **Automatic** -->drop-down arrow
 - Select **(Gantt bar)**
6. Click **Analysis** -----> one dialog box appears
 - Select **CREATE CALCULATED FIELD**
 - **Add Name:** OrderUntilShip
 - **Add Formula:** DATEDIFF('day', [Order Date],[Ship Date])
7. After adding Name and Formula apply and click ok
8. Drag **OrderUntilShip** into size on Marks card
9. Click **SUM (OrderUntilShip)** drop-down arrow
10. One dialog box appears in that select Measure(sum) and click Average
11. Drag **Ship Mode** to color area in marks card

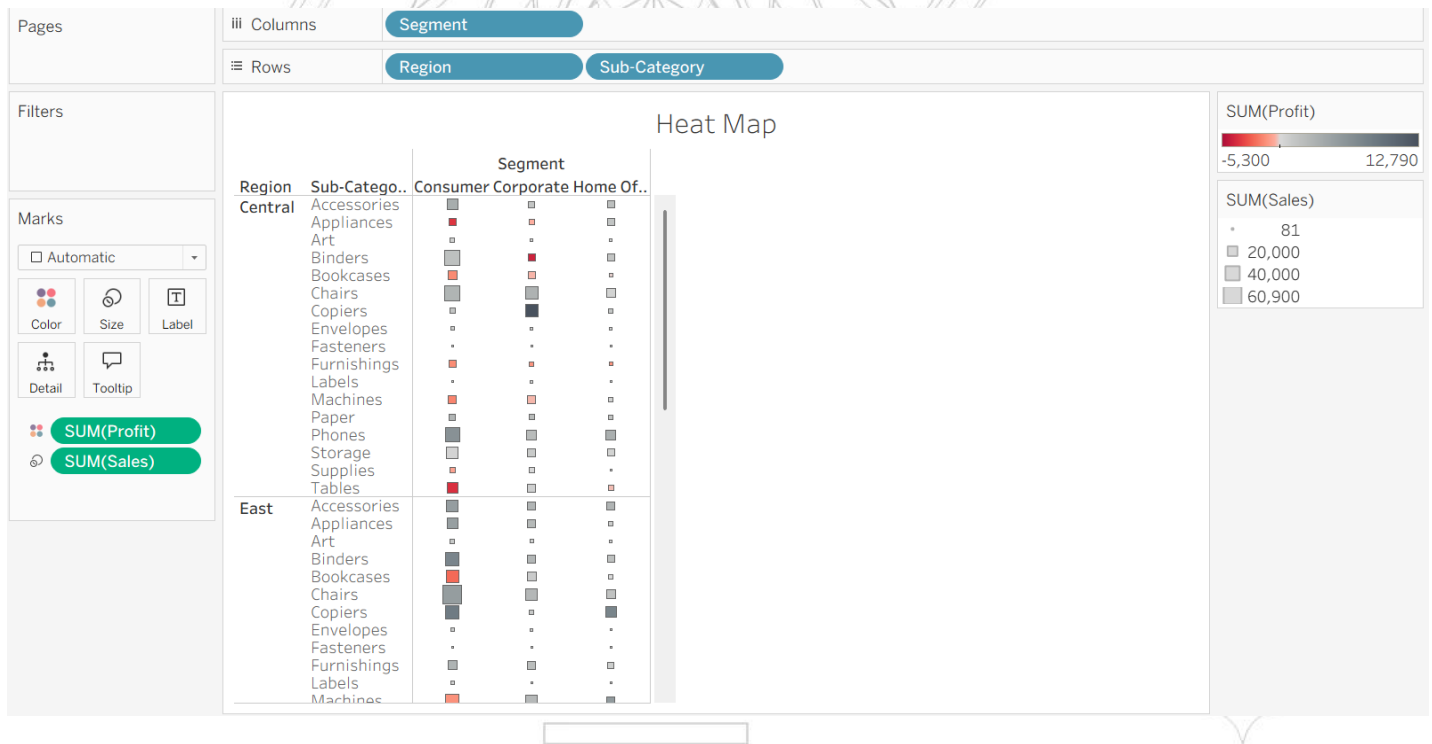
Output:



Heat Map

1. Drag **[segment]** placed into columns shelf
2. Drag **[Region]** into rows shelf
3. Drag **[sub-category]** into rows shelf and placed into right side of region
4. Move **profit** to the Marks card and drop it into colour
 - Click on color area and one dialog box will appear
 - Click **Edit colour option**
 - Change palette Click drop down **Automatic** arrow
 - Select **Red-black-Diverging**
 - Click use full color range
5. Place **sales** into size
6. Click on size in Mark card adjust size to right

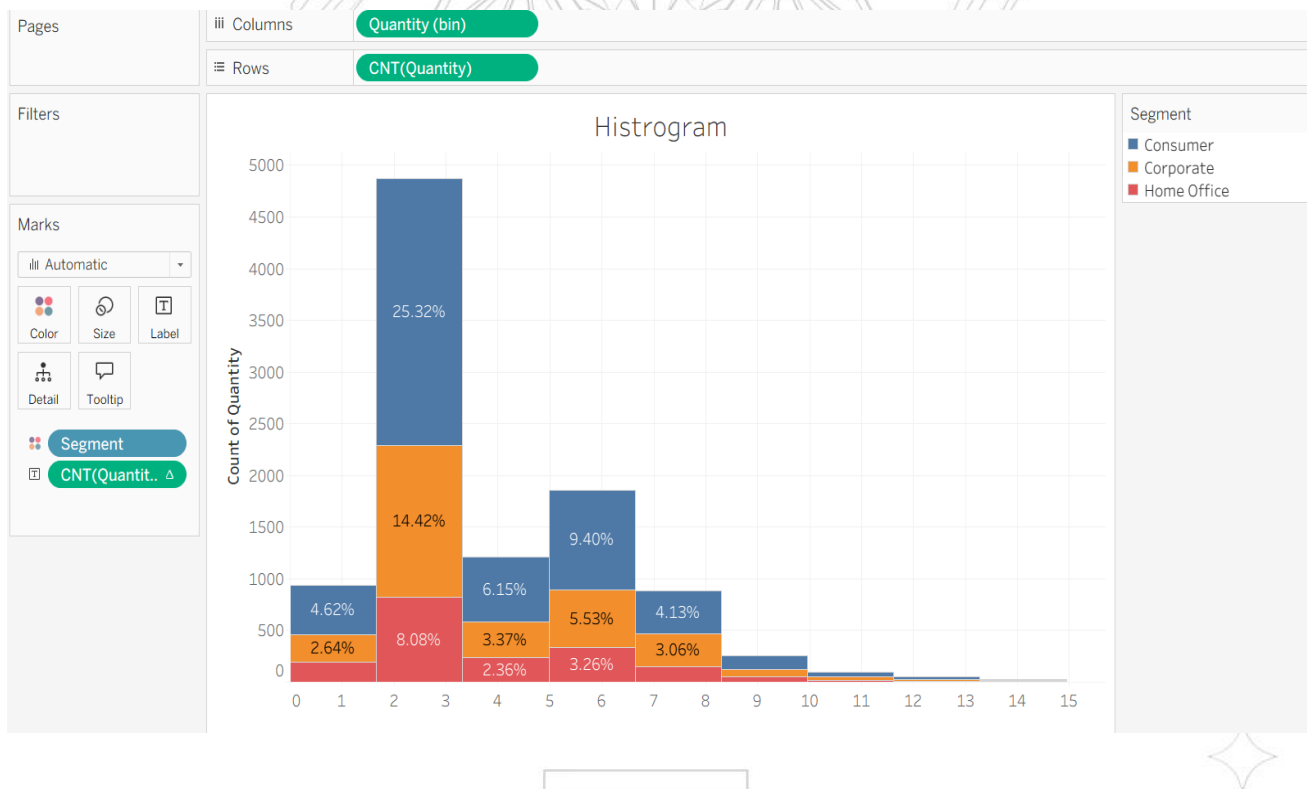
Output:



Histogram

1. Move **[Quantity]** into Columns shelf
2. Click show me in toolbar choose **Histogram**
3. It creates bin automatically
4. In Marks card add **[Segment]** to colour
5. **[CNT quantity]** default appears in Rows shelf
6. In rows shelf copy **CNT quantity** to label using **CTRL drag**
7. Right click on **CNT quantity** on Marks
8. Select **Quick table calculation** and click **percent of total**

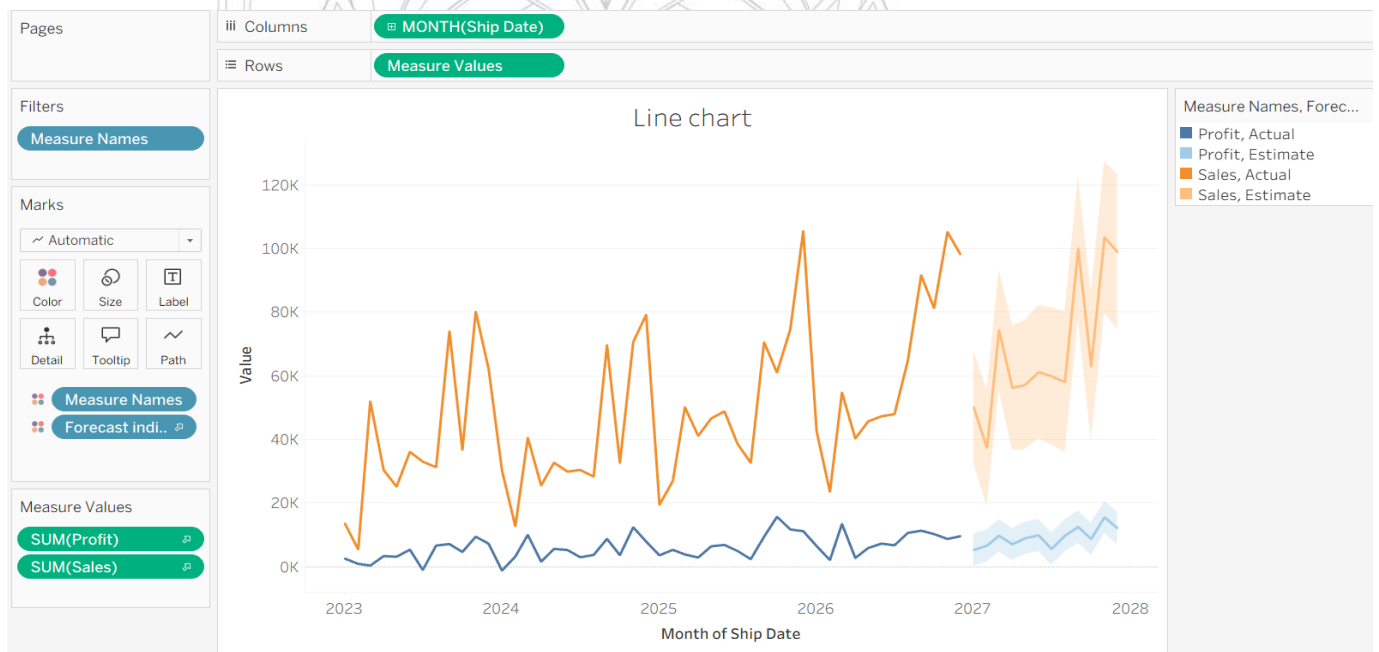
Output:



Line chart

1. Drag **[Ship date]** placed into Columns shelf
2. Drag **[Sales]** placed into Rows shelf
3. Drag **[Profit]** into rows shelf placed into right side of **sales**
4. Now we see two separate axes
 - Sales axis
 - Profit Axis
5. Drag **[SUM(Profit)]** from Rows shelf and directly drop on the **Sales axis**
6. Click on **Analytics**
7. Click and drag **Forecast** and drop into a worksheet area
8. In columns shelf right click on **[YEAR (Ship date)]** change year to Month

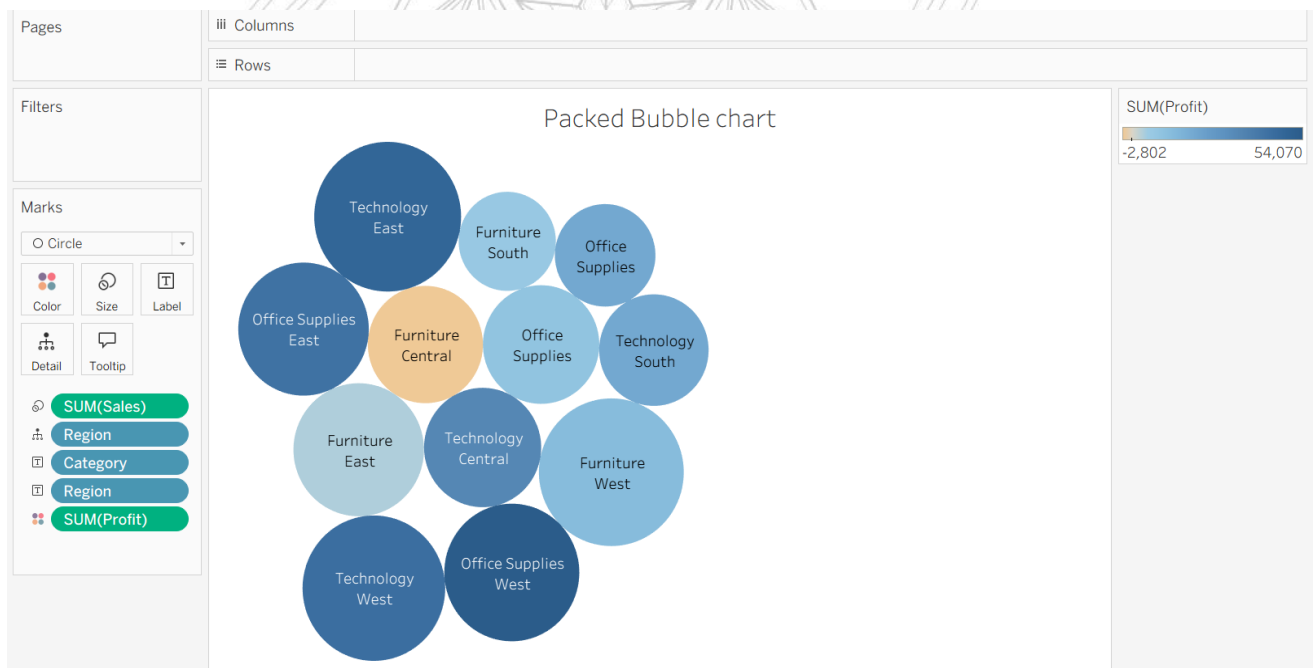
Output:



Packed Bubble Chart

1. Drag **[category]** placed in a columns shelf
2. Drag **[sales]** into rows shelf
3. Click on show me
4. Select **packed bubble chart**
5. Drag **[Region]** and drop it into detail in Marks card
6. Drag **[profit]** into color in Marks card
7. Again, drag **[Region]** and drop it into Label

Output:



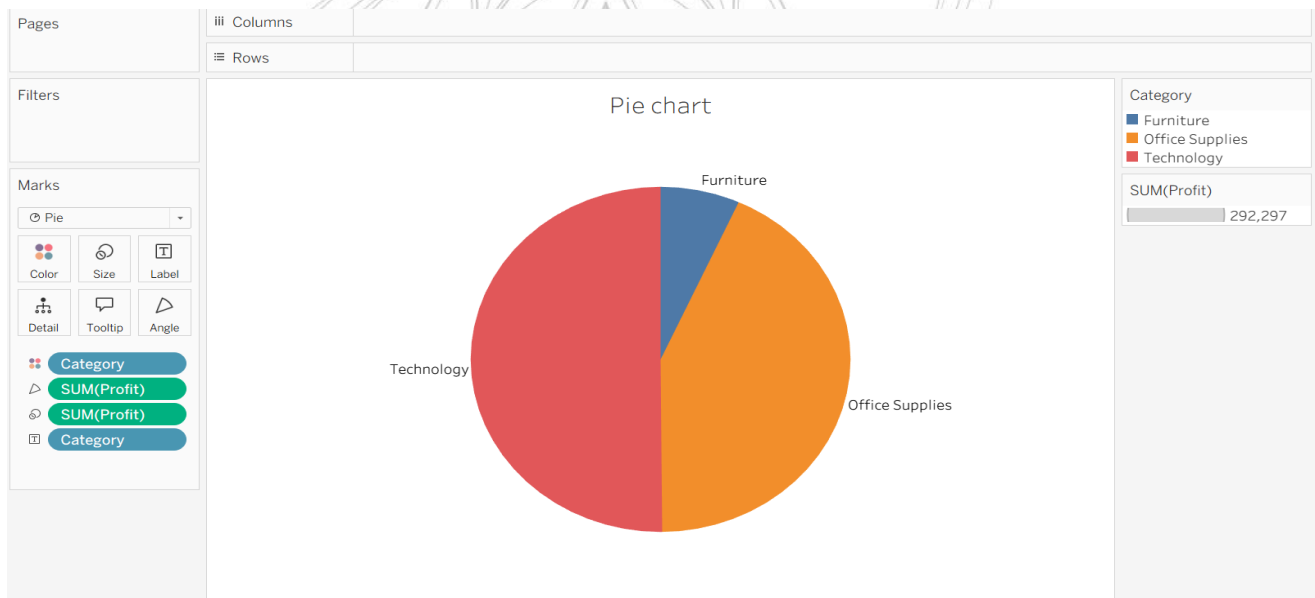
— INNOVATORS —



Pie chart

1. Drag **[profit]** placed into Columns shelf
2. Drag **[category]** into Rows shelf
3. Click on show me and select **pie chart**
4. In toolbar **standard Menu** appears
5. In Standard menu click drop down arrow and select **Entire view**
6. Drag **[category]** placed into **label** in Marks card

Output:



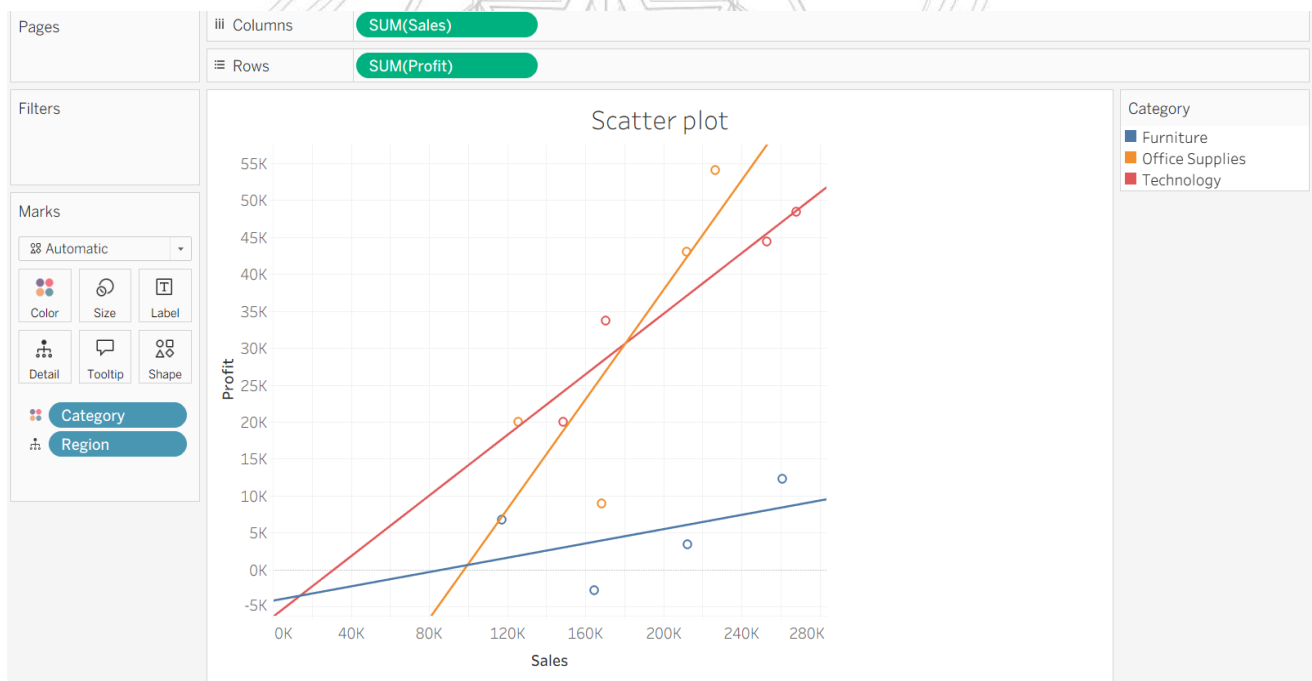
ALPHA
- INNOVATORS -



Scatter plot

1. Drag **[sales]** placed into columns shelf
2. Drag **[profit]** into Rows shelf
3. Move **category** and drop it on colour in Marks Card
4. Drag **Region** and drop it into detail in Marks Card
5. Click on **Analytics**
6. Click and drag **Trend line** and drop down into a worksheet area

Output:



— INNOVATORS —



Text Table with Percentages

1. Move **[Order Date]** placed into Columns shelf
2. Add **[Sub-Category]** into Rows shelf
3. Drag **[sales]** and add into a **Text** in marks
4. Drag **Region** placed into left side of **Sub-Category** in Rows shelf
5. In marks card right click on **[SUM (sales)]**
 - Select Add Table calculation
 - **Choose Calculation type:** Percent of total
 - **Computer Using:** Pane down

Output:

Pages

Columns: YEAR(Order Date)

Rows: Region, Sub-Category

Filters

Marks: Automatic, Color, Size, Text, Detail, Tooltip, SUM(Sales)

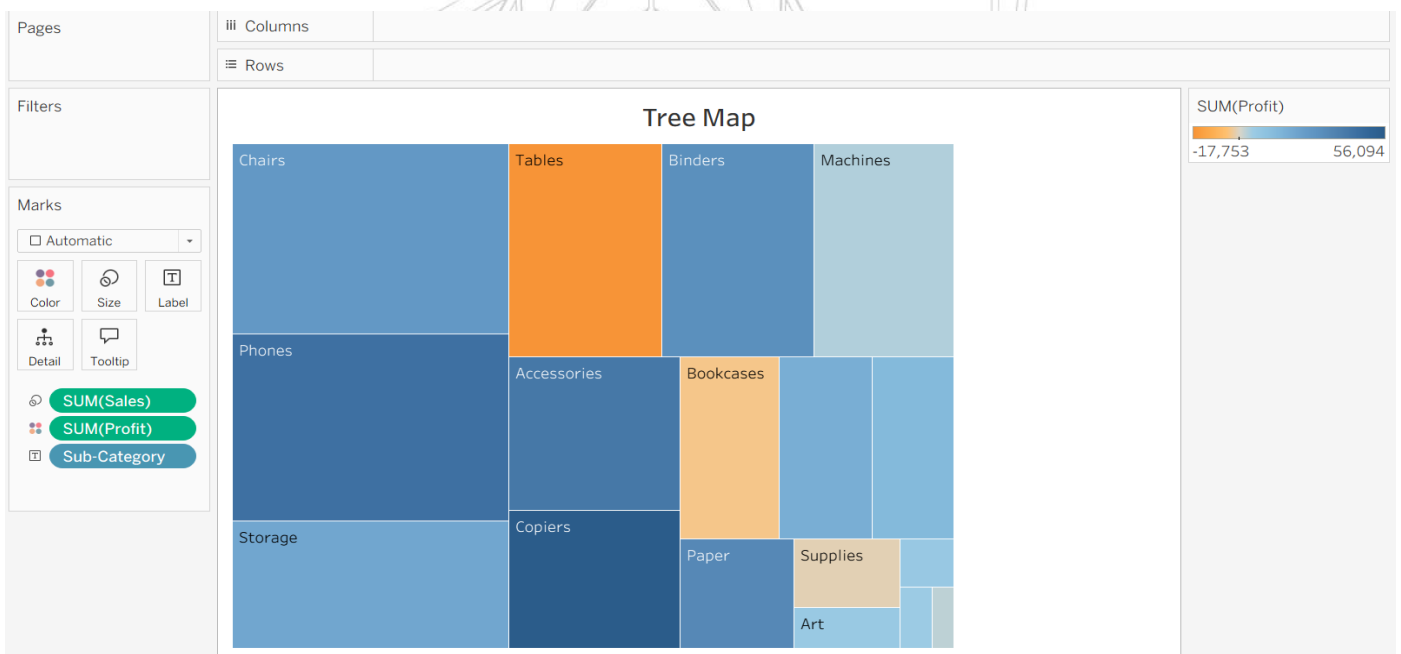
Text Table with percentage

Region	Sub-Category	2023	2024	2025	2026
Central	Accessories	4.27%	7.58%	7.27%	7.39%
	Appliances	3.52%	4.84%	4.05%	6.04%
	Art	0.79%	1.10%	1.02%	1.55%
	Binders	15.28%	5.73%	10.19%	14.31%
	Bookcases	1.77%	8.07%	5.64%	3.81%
	Chairs	19.99%	17.41%	15.74%	16.15%
	Copiers	3.15%	12.45%	11.78%	2.49%
	Envelopes	1.54%	0.85%	0.65%	0.81%
	Fasteners	0.12%	0.09%	0.17%	0.22%
	Furnishings	2.44%	2.46%	3.45%	3.44%
	Labels	1.01%	0.30%	0.34%	0.40%
	Machines	15.69%	1.80%	1.79%	4.05%
	Paper	2.26%	3.45%	3.61%	4.22%
	Phones	9.56%	18.82%	13.39%	15.70%
East	Storage	10.68%	8.10%	8.63%	9.28%
	Supplies	0.42%	0.32%	2.89%	2.98%
	Tables	7.50%	6.67%	9.37%	7.16%
	Accessories	4.47%	11.40%	3.41%	6.85%
	Appliances	4.53%	4.26%	5.22%	5.73%
	Art	1.01%	1.13%	1.11%	1.27%
	Binders	5.40%	9.18%	10.81%	6.84%

Tree Map

1. Drag **[Sub-Category]** Placed into a columns shelf
2. Drag **[sales]** into rows shelf
3. Click show me and select Tree map
4. Move **Ship Mode** and drop it into color in Marks card
5. Drag **profit** add into color in Marks card

Output:



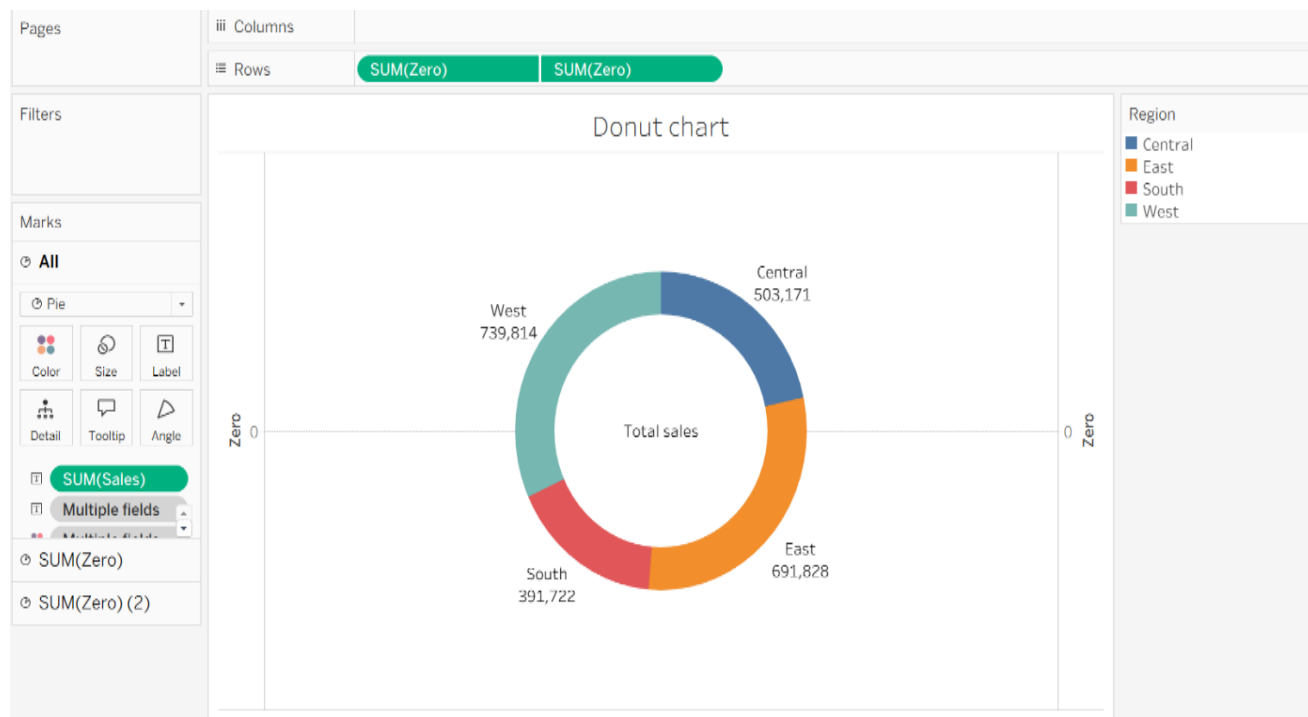
ALPHA
- INNOVATORS -



Donut Chart

1. Open the **[Analysis menu]** and choose **Create Calculated Field**
2. Enter the **[Name as Zero]** and type the **[value 0]** then press OK
3. From data drag **Zero** and placed into the Rows shelf
4. In the Marks card, change the chart type from **Automatic to Pie**
5. In tool bar click on standard drop down arrow
6. Then click the **Entire View**
7. Adjust the Size slider in the Marks card to enlarge the pie chart
8. Add **Zero** once again to the Rows shelf beside the **first SUM(Zero)** field
9. Right-click the **[Second SUM(Zero)]** and apply the **Dual Axis option**
10. Select the **Second SUM(Zero)** in Marks card
11. Reduce the chart size slightly and choose **White as the color**
12. In Marks card select **SUM (zero)**
13. Drag the **Region** into the Color section of the Marks card
14. Again, place **Region** into the Label area
15. Move the **Sales** into the Label shelf
16. Drag **Sales** to the Angle option in the Marks card
17. Click on the **Second SUM(Zero)** Marks card
18. Add **Sales** to the Label section once more
19. Open the Label settings
 - click the **Three-dot Text Editor Option**
 - Type **Total Sales**, and confirm by clicking OK

Output:

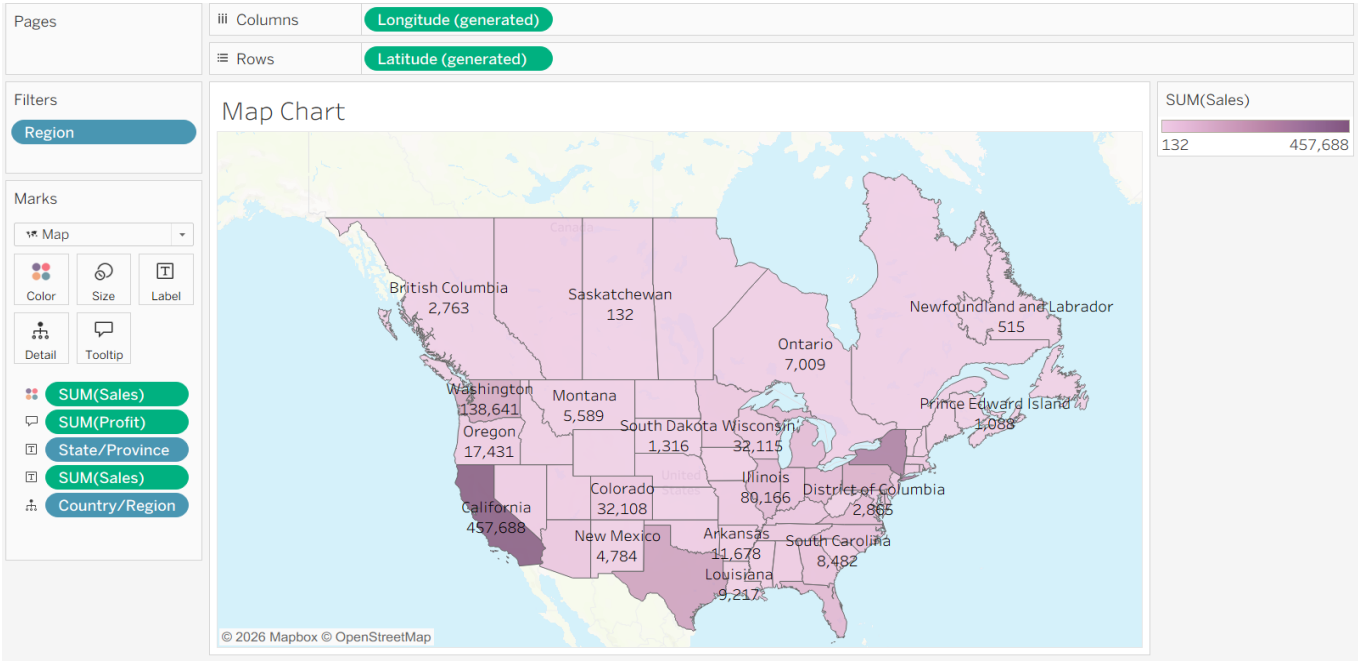


Map Chart

1. Drag **[State/Country]** from the Data Pane to the workspace.
 - Tableau automatically creates a map using Latitude and Longitude.
2. Drag **[Sales]** to the Color option in the Marks card.
3. In the Marks card, click the drop-down menu.
 - Choose either **[Map or Filled Map]**.
4. Click on Color in the Marks card.
 - Select **[Edit Colors]**.
 - In palette click **Automatic** drop-down arrow and select purple
 - Click OK to apply
5. Drag **[State]** to the Label shelf.
6. Drag **[Sales]** to the Label shelf.
7. Drag **[Region]** to the Filters shelf.
 - Select the choose all regions.
 - Click OK to apply the filter.
8. Drag **[Profit]** to the Tooltip in Marks card
9. In tool bar go to **[Map Menu]**
 - Click **[Background style]**
 - Select **[outdoors]** 



Output:



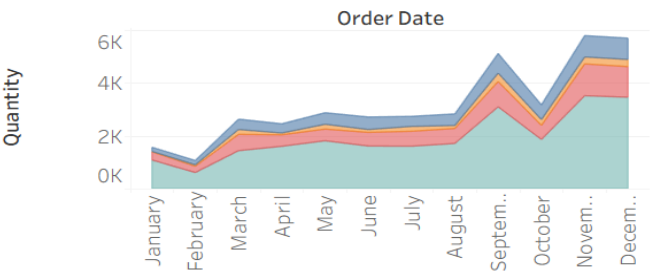
ALPHA
— INNOVATORS —



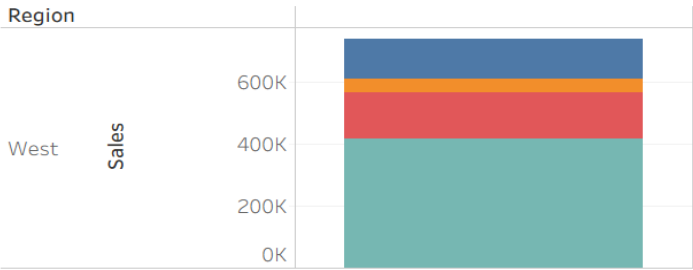
DASHBOARD

Dashboard1

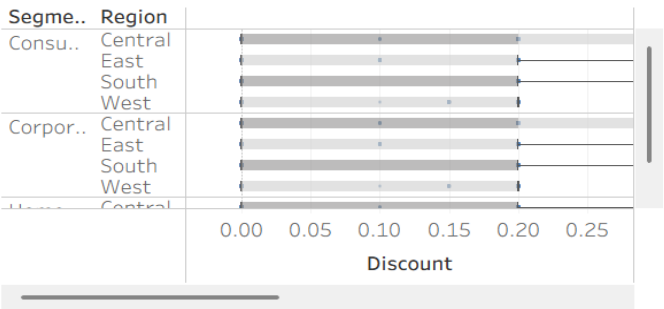
Area chart



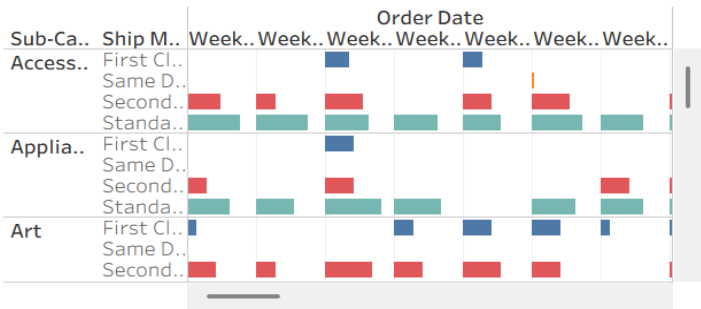
Bar chart



Box plot



Gantt chart



ALPHA
- INNOVATORS -

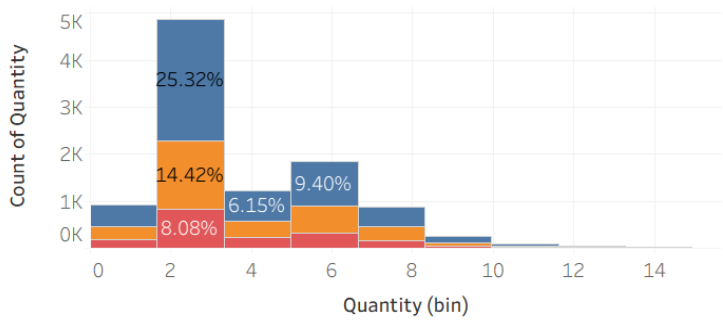


Dashboard2

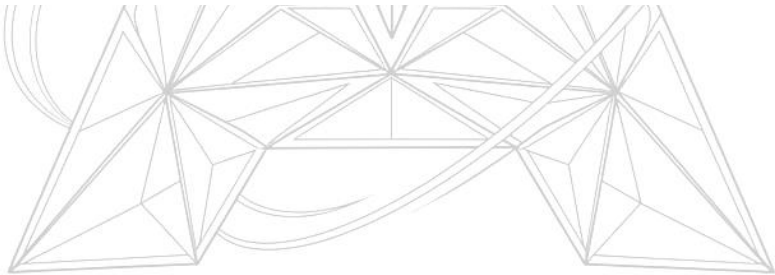
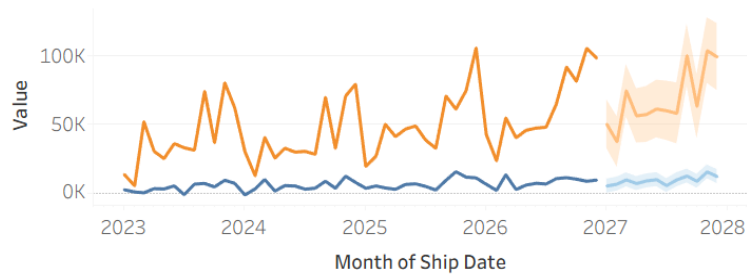
Heat Map

Region	Sub-Catego..	Segment		
		Consu..	Corpor..	Home O..
Central	Accessories	■	■	■
	Appliances	■	■	■
	Art	■	■	■
	Binders	■	■	■
	Bookcases	■	■	■
	Chairs	■	■	■
	Copiers	■	■	■
	Envelopes	■	■	■
	Fasteners	■	■	■
	Furnishings	■	■	■
	Labels	■	■	■
	Machines	■	■	■
	Paper	■	■	■
	Phones	■	■	■
	Storage	■	■	■
East	Supplies	■	■	■
	Tables	■	■	■
	Accessories	■	■	■
	Appliances	■	■	■
	Art	■	■	■
	Binders	■	■	■
	Bookcases	■	■	■
	Chairs	■	■	■
	Copiers	■	■	■
	Envelopes	■	■	■
	Fasteners	■	■	■
	Furnishings	■	■	■
	Labels	■	■	■
	Machines	■	■	■
	Paper	■	■	■
	Phones	■	■	■

Histogram



Line chart



ALPHA
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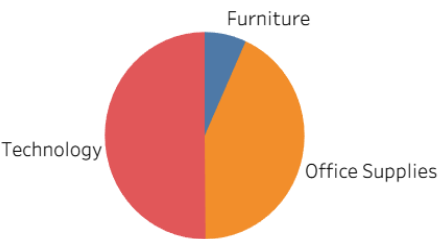


Dashboard3

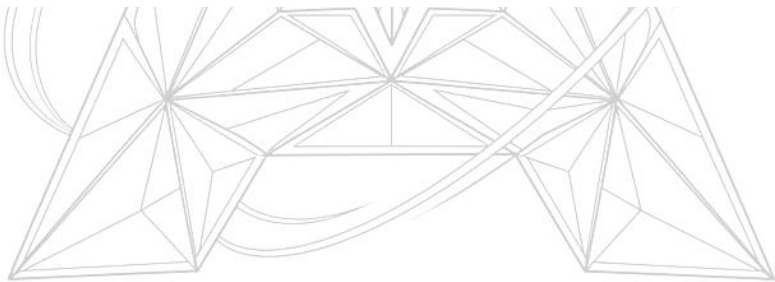
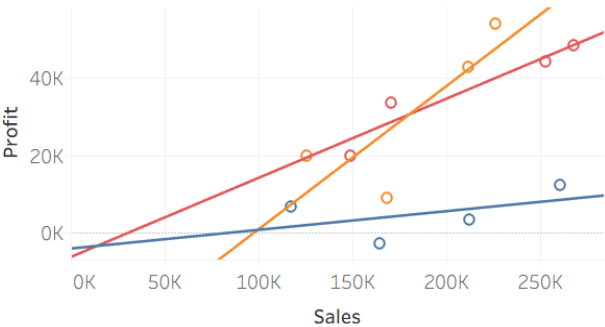
Packed bubble chart



Pie chart



Scatter plot



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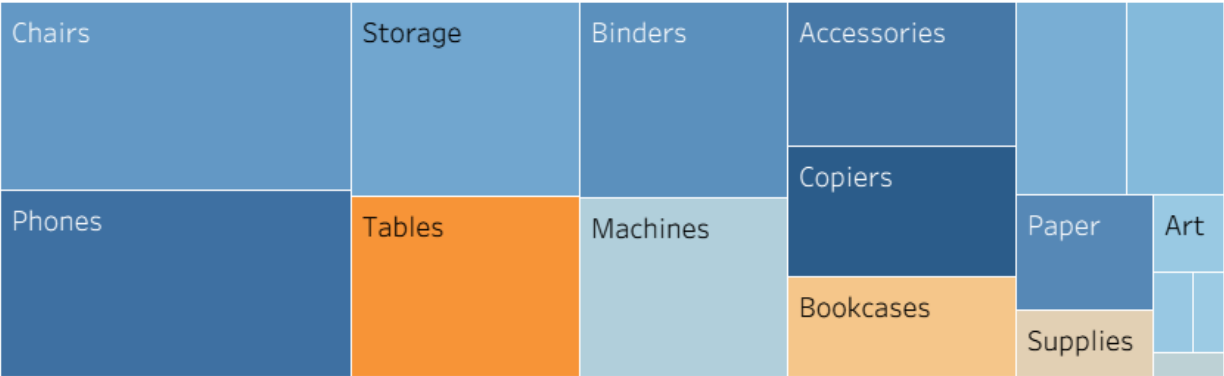


Dashboard4

Text Table with percentages

Region	Sub-Catego..	Order Date			
		2023	2024	2025	2026
Central	Accessories	4.27%	7.58%	7.27%	7.39%
	Appliances	3.52%	4.84%	4.05%	6.04%
	Art	0.79%	1.10%	1.02%	1.55%
	Binders	15.28%	5.73%	10.19%	14.31%
	Bookcases	1.77%	8.07%	5.64%	3.81%
	Chairs	19.99%	17.41%	15.74%	16.15%
	Copiers	3.15%	12.45%	11.78%	2.49%
	Envelopes	1.54%	0.85%	0.65%	0.81%
	Fasteners	0.12%	0.09%	0.17%	0.22%

Tree Map

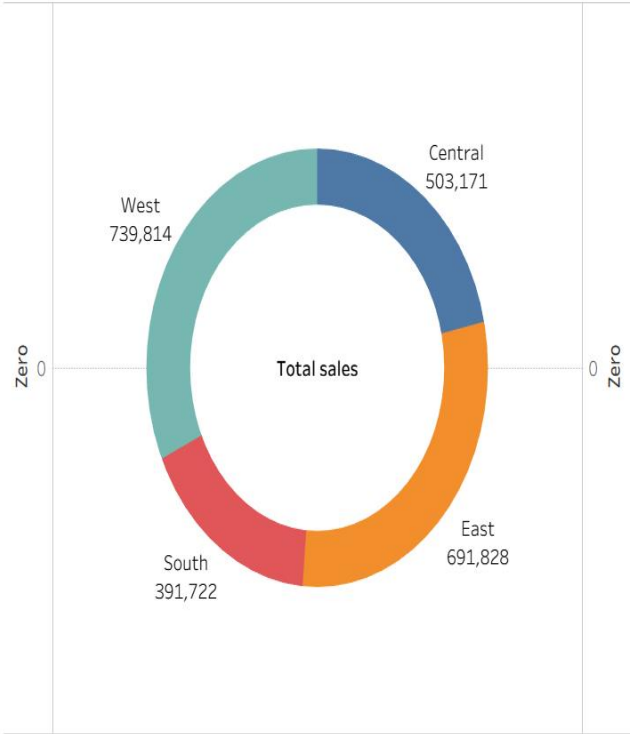


ALPHA
- INNOVATORS -

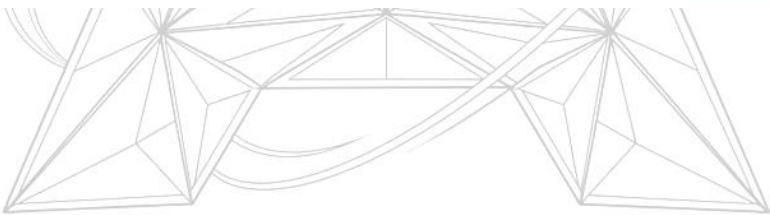
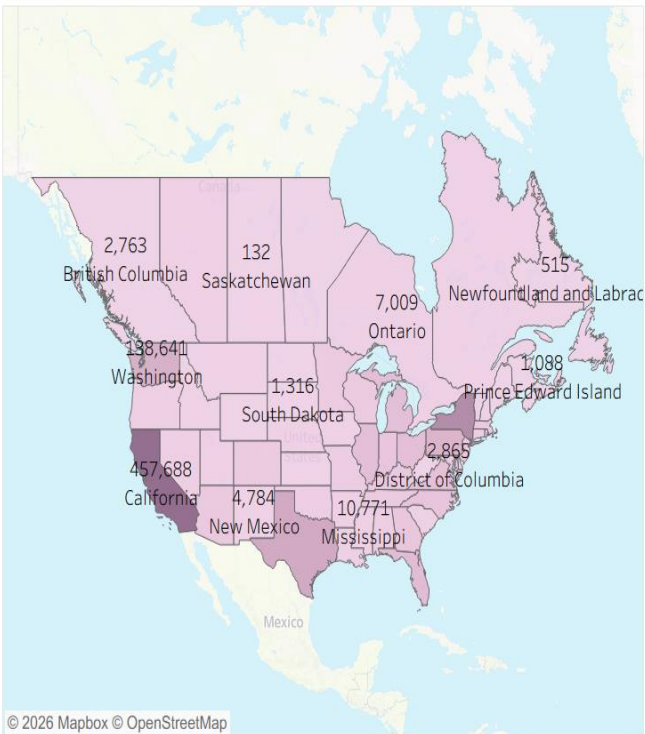


Dashboard5

Donut chart



Map chart



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